CAUÃ FELYPE CAETANO REIS

Junior Python Developer · Asynchronous Systems, FastAPI & Clean Code

📍 Goiás, Brazil (Remote) 📞 +55 (62) 9 9452-1953 ✉ caua.reis.dev@gmail.com 🔗 linkedin.com/in/cauafelypereis
💻 github.com/cauacreis 🌐 Portfolio (Vercel)

PROFESSIONAL SUMMARY

Analytical and results-driven Junior Python Developer with 2+ years of hands-on software development experience and a proven track record of architecting, deploying, and maintaining high-availability asynchronous backend systems in production. Lead architect and operator of **SKZ Trading**, a 24/7 digital asset arbitrage SaaS orchestrating 11 distributed Python microservices (`asyncio` , `aiohttp` , FastAPI) maintaining 99.9% uptime with sub-100ms latency and zero transactional loss. Co-author of an applied AI and software engineering research paper submitted to the prestigious **IEEE Transactions on Software Engineering (IEEE TSE)**. Solid foundation in object-oriented programming, data structures, algorithms, and relational databases. Certified in Agile (**Scrum SFC™**), advanced English speaker prepared for global client collaboration, and driven by continuous learning, code quality, and peer review best practices.

CORE TECHNICAL SKILLS

Python & Backend Engineering
Python 3.11+, AsyncIO, aiohttp, FastAPI, RESTful API design, Pydantic v2, Object-Oriented Programming (OOP), Data Structures & Algorithms, Design Patterns.

Microservices & Background Workers
Asynchronous Task Queues, Worker Daemons, Event Loops, WebSockets (real-time bidirectionality), Microservices Architecture, Process Management (PM2, systemd).

Databases & Data Modeling
PostgreSQL, MySQL, SQLite (Outbox pattern), Redis (in-memory caching & pub/sub), relational schema design, query optimization, indexing, ACID transactions.

DevOps, Testing & Agile
Git & GitHub (GitFlow, PRs, Code Reviews), Docker, Linux/VPS, Pytest, Unit Testing, CI/CD (GitHub Actions), Clean Code, Scrum (SFC™ Certified).

PROFESSIONAL EXPERIENCE & PRODUCTION SYSTEMS

Python Software Engineer & Lead Architect — SKZ Trading `24/7 Production` Jul 2026 – Present

High-throughput digital asset algorithmic trading SaaS integrated with real-time financial and partner APIs

- **Distributed Python Microservices:** Architected, deployed, and maintain 11 asynchronous microservices in Python (`asyncio` , `aiohttp`) running continuous 20-second market scans on Linux VPS with sub-100ms API response latency.
- **FastAPI Endpoint Architecture:** Built and documented RESTful API endpoints using FastAPI and Pydantic v2 schemas for robust serialization, parameter validation, and error handling.
- **Algorithmic Pricing Engine:** Engineered a proprietary liquidity evaluation algorithm with liquidity scoring, minimum 12% buy discount, and hard-coded anti-loss circuit breakers preventing sales below acquisition costs.
- **Financial Pipelines & Webhooks:** Modeled relational MySQL databases with ACID compliance, automating split payouts via Mercado Pago webhooks (PIX) and maintaining 99.9% uptime across 200+ completed trades.
- **Security Hardening:** Implemented server-side encryption (Ed25519/AES) adhering to OWASP Top 10 guidelines, eliminating exposed secrets and securing API payloads in transit.

APPLIED ENGINEERING PROJECTS & RESEARCH

OrenAI — Enterprise AI Platform & Research Paper (IEEE TSE) Python / Machine Learning 2025 – Present

- **Machine Learning & XAI Pipelines:** Co-developed predictive models combining Gradient Boosting (XGBoost/LightGBM) with Explainable AI (SHAP), enabling complete mathematical transparency for geospatial mineral exploration targets.
- **Multi-Agent Architecture:** Engineered containerized agent workflows in Docker to isolate commodity processing pipelines, accelerating spatial computation and benchmarking.
- **Scientific Publication:** Co-authored a research paper submitted to the prestigious **IEEE Transactions on Software Engineering (IEEE TSE)**, responsible for system topology diagrams, mathematical formalization, and technical documentation.

LeadProspector & LinkedinTalk — Autonomous Python Agents Python 3.11 / Redis / Postgres 2025 – 2026

- **State Machine Automation:** Built an autonomous B2B lead ingestion pipeline in Python 3.11 with Docker Compose, Redis, and PostgreSQL, implementing an 8-stage state machine and automated outreach flows.
- **Defensive API Architecture:** Built a companion personal branding copilot in Python (FastAPI) and TypeScript featuring anti-rate-limit algorithms, automated unit tests, and GitHub Actions CI/CD.

Bodega do Vidigal — Local-First POS & Kitchen Display System (KDS) Systems & Relational Persistence 2026

- **High-Resilience Outbox Pattern:** Modeled local SQLite relational persistence with an atomic Outbox pattern, guaranteeing zero duplicate orders and complete state recovery upon unexpected restarts.
- **Automated Testing Suite:** Implemented 37 automated tests (Jest) validating order state machines, FIFO kitchen prioritization, and nightly automatic reset routines.

EDUCATION & CERTIFICATIONS

B.S. in Software Engineering — UniEVANGÉLICA (Anápolis, GO) 4th Semester | Expected: 2027/2028

- Evening classes — **100% available for full-time business hours (Monday – Friday)**.
- Vice-President of the CodeUp Academic Directory; Director of Arts & Tech Culture.

Desktop Developer Track: C#, Java, Relational Databases (SQL), OOP, UML — Instituto Mix Oct 2022 – Jun 2024

- Practical coursework in Object-Oriented Programming, Relational Modeling, and Distributed System Fundamentals.

- **Certifications:** Scrum Fundamentals Certified (SFC™) · Technical Storytelling & System Documentation.
- **Languages:** Portuguese (Native) · English (Advanced — Fluent technical documentation, scientific writing, and verbal collaboration).